Year 2 Statistics and Mechanics

Chapter 4

Moments

Practice Booklet

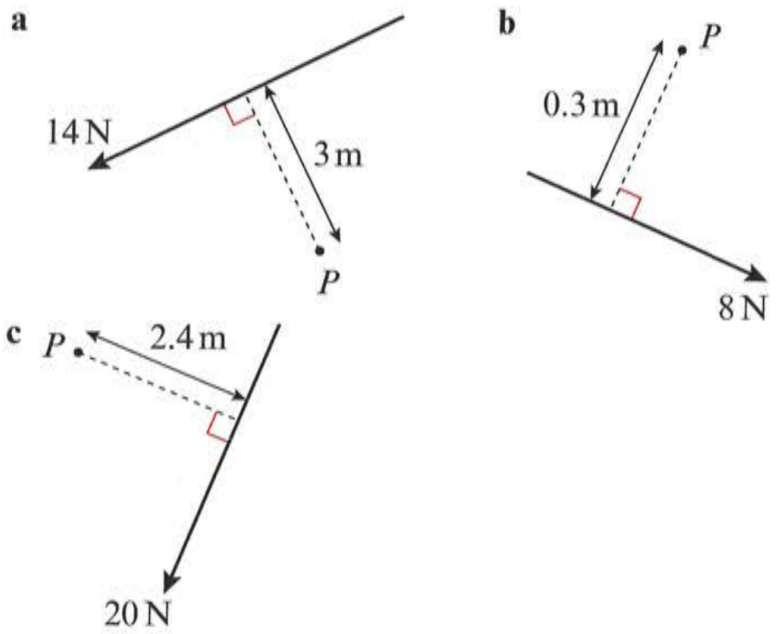
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4.1 Moments

Practice Book, printed page 33.

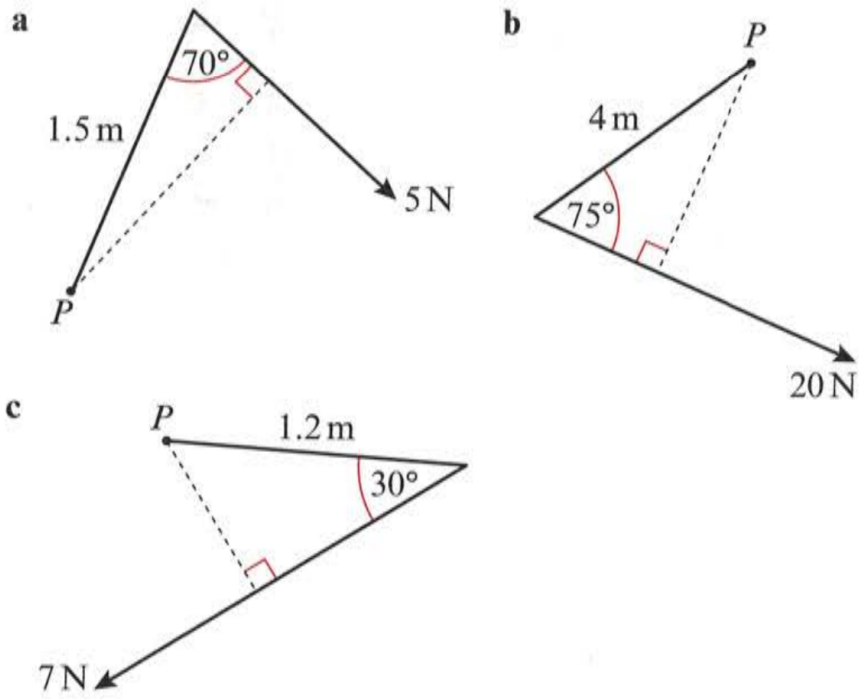
1. Find the moment of the force acting about the point P.



Hint

You should include the direction of rotation when you describe the moment of the force.

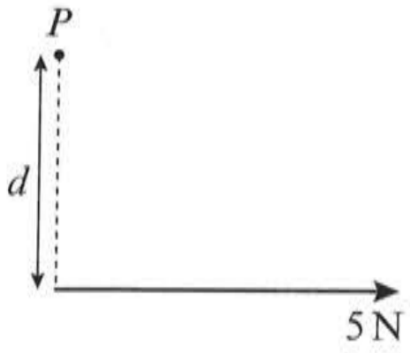
2. Find the moment of the force acting about the point P.



Hint

Multiply the magnitude of the force by the perpendicular distance from the axis of rotation.

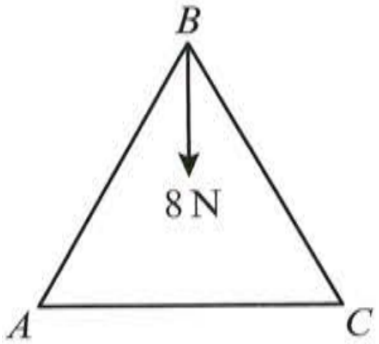
3. A force of 5 N acts about the point P as shown in the diagram.
Given that the force produces a moment of 1.5 Nm in the anticlockwise direction, work out the distance d.



4. ABC is a lamina in the shape of an equilateral triangle of side 20 cm. A force of 8 N acts vertically downwards at B, as shown in the diagram.
Work out the moment of this force about:

a. A

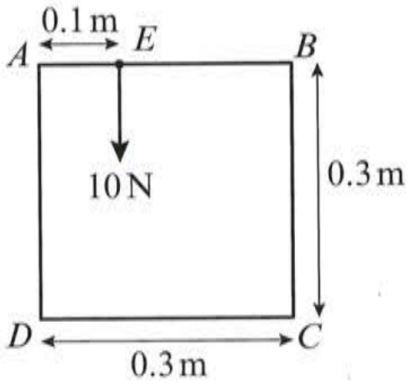
b. C



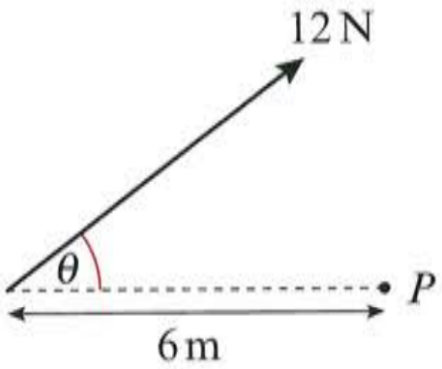
5. $ABCD$ is a lamina in the shape of a square of side 0.3 m. A force of 10 N acts vertically downwards at the point E that lies on the line AB, 0.1 m from A.
Work out the moment of this force about:

a. A

b. C



6. A force of 12 N produces a moment of 3600 N cm in the clockwise direction about the point P as shown in the diagram. Work out the size of angle θ.

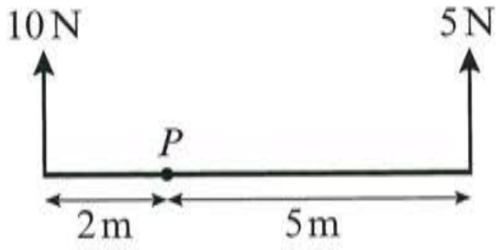


7. E/P A mass hangs from a horizontal wooden beam AB of length 0.6 m. The mass is fixed at a point C which lies on the beam between A and B. Given that the moment of the weight about A is twice the moment about B, work out the length AC. (4 marks)

4.2 Resultant moments

Practice Book, printed page 34.

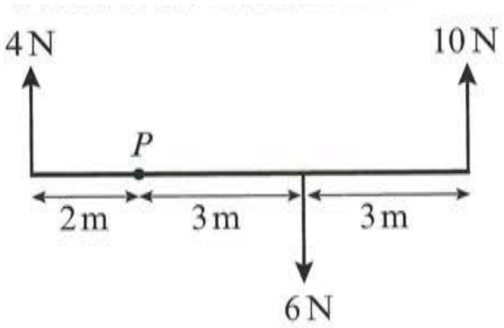
1. The diagram shows the forces acting on a light rod.
Work out the resultant moment acting about the point P.



Hint

Find the anticlockwise moment caused by the 5 N force and subtract the clockwise moment caused by the 10 N force.

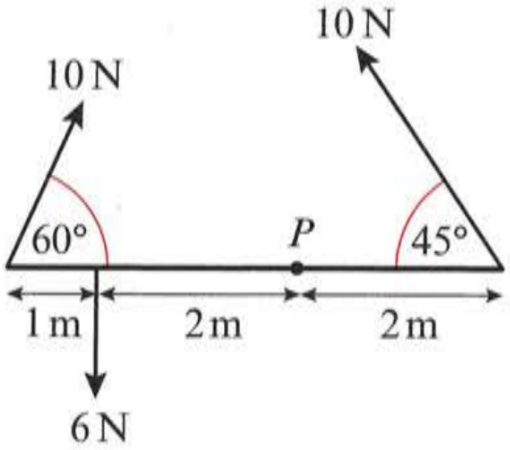
2. The diagram shows the forces acting on a light rod.
Work out the resultant moment acting about the point P.



Hint

Set clockwise as the positive direction, so that anticlockwise moments will be negative. Then add all the moments together. If the answer is negative, the resultant moment will act in the anticlockwise direction.

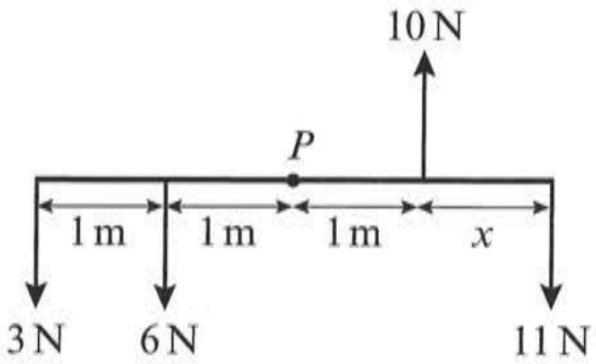
3. The diagram shows the forces acting on a light rod. Calculate the resultant moment acting about the point P.



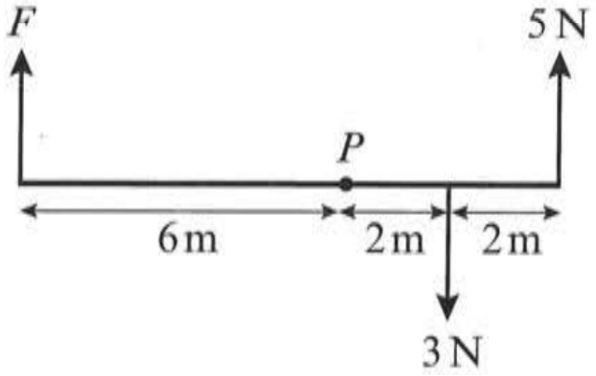
Hint

First work out the perpendicular distance of each force from P.

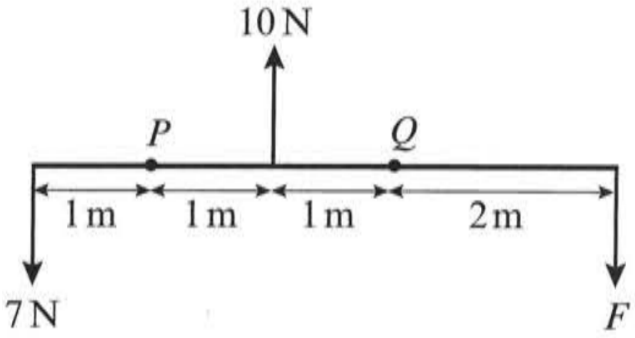
4. The diagram shows the forces acting on a light rod. Given that the resultant moment acting about P is 0, work out the length x.



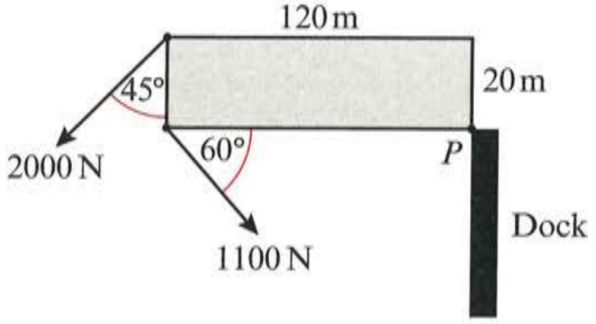
5. E The diagram shows the forces acting on a light rod. Given that the resultant moment acting about P is 5 Nm in the clockwise direction, work out the value of F. (4 marks)



6. E The diagram shows the forces acting on a light rod. Given that the resultant moment acting about Q is twice the resultant moment acting about P, and that both act in the clockwise direction, work out the value of F. (3 marks)



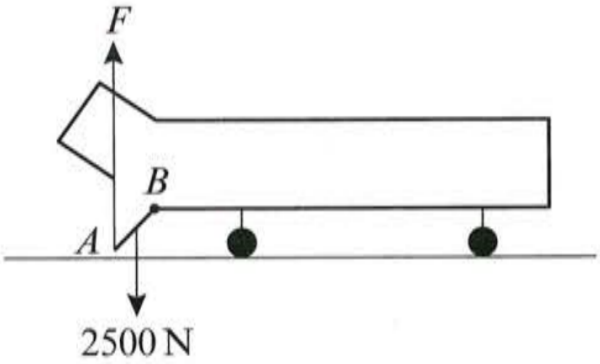
7. E/P A barge is tethered to a dock at the point P and is being moved into position by two tugs. The forces exerted by the two tugs are shown in the diagram.



a. Calculate the resultant moment acting about P. (4 marks)

b. State one modelling assumption that you have used. (1 mark)

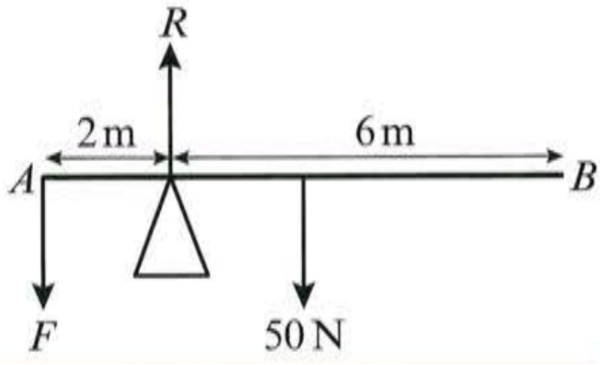
8. E/P The loading door, AB, on a transport aeroplane is to be closed. The door is 4 m long and has a weight of 2500 N. The door is angled at 45° to the horizontal and hinged at B. A force of F N acts vertically upwards at A, as shown in the diagram.
By modelling the door as a uniform rod, show that in order for the door to move, F must be greater than 1250 N. (3 marks)



4.3 Equilibrium

Practice Book, printed page 36.

1. A uniform rod of length 8 m and weight 50 N rests on a support that is 2 m from A. The rod is held in a horizontal position by a force F that acts vertically downwards at A.



a. Work out the magnitude of F.

Hint

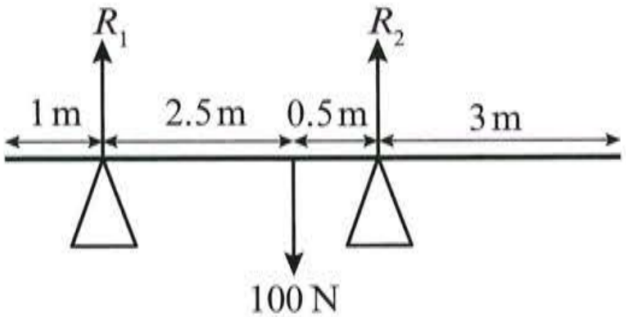
The rod is in equilibrium so the resultant moment is zero. This means the sum of the clockwise moments equals the sum of the anticlockwise moments.

b. Work out the reaction at the support.

Hint

Resolve vertically to obtain $R = F + 50$.

2. A uniform rod of length 7 m and weight 100 N rests horizontally on two supports.

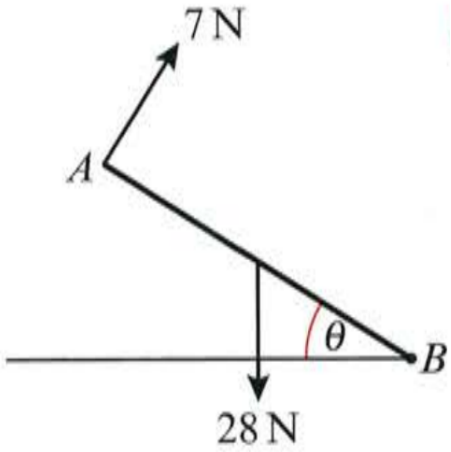


Work out the values of the reactions R_1 and R_2 acting at the supports.

Hint

Take moments about the point that makes the calculations as simple as possible.

3. A uniform rod AB, of length 2 m and weight 28 N, is hinged at B, and is held in equilibrium by a force of 7 N acting perpendicularly to the rod at A. The rod makes an angle of θ with the horizontal, as shown in the diagram. Find the size of angle θ.

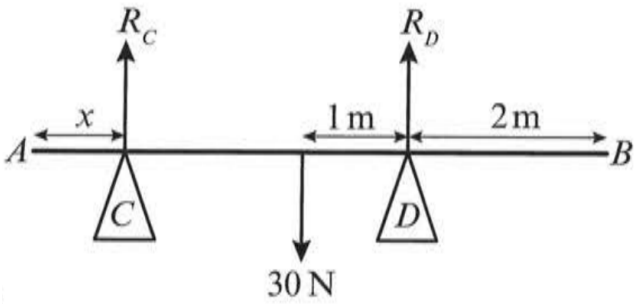


Hint

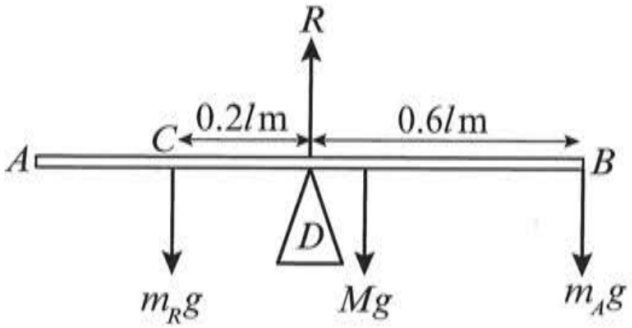
By taking moments about B you can ignore the reaction at the hinge.

4. P A uniform rod AB of mass m kg and length l m, is held horizontally in equilibrium by two vertical strings attached at points A and C where $AC : CB = 3 : 1$. Find, in terms of m, the tensions T_A and T_C in the strings attached to the rod at A and C respectively.

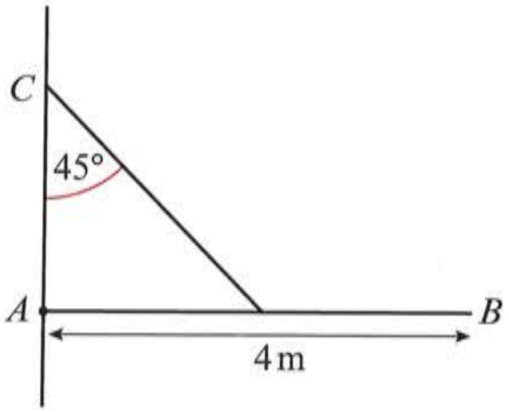
5. E/P A uniform rod AB, of weight 30 N and length 6 m, sits horizontally in equilibrium on two supports at points C and D as shown in the diagram.
The reactions of the supports on the rod at points C and D are R_C and R_D respectively.
Given that $R_C = 4R_D$, work out the length x. (4 marks)



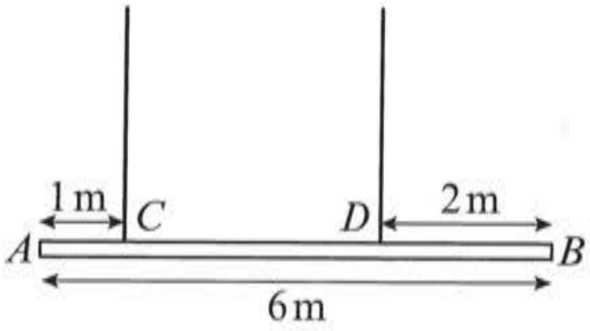
6. E/P A seesaw is modelled as a uniform rod AB of mass M kg and length l m. It rests on a support at D where $AD = 0.4l$ m. Ruairi, who has mass m_R kg, sits $0.2l$ m from the support at point C and Athena, who has mass m_A kg, sits at B as shown in the diagram.
Given that the seesaw rests horizontally in equilibrium, show that $M = 2(m_R - 3m_A)$. (3 marks)



7. E/P A rod AB, of length 4 m and mass 10 kg, is hinged to a vertical wall at A. It is held in the horizontal position by a rope that is attached at one end to the midpoint of the rod and at the other end to the point C that lies vertically above A. The rope makes an angle of 45° with the vertical, as shown on the diagram.
Work out the tension in the rope. (3 marks)



8. E/P A plank AB has mass 10 kg and length 6 m. It is held in equilibrium in a horizontal position by two ropes attached at the points C and D, where $AC = 1$ m and $DB = 2$ m.
The plank can be modelled as a uniform rod and the ropes as light strings.



a. Work out:

i. the tension in the rope at C

ii. the tension in the rope at D.

(4 marks)

A mass of 5 kg is placed on the plank at a distance x m from C, between C and D. The mass can be modelled as a particle.
Given that the plank remains in equilibrium in the horizontal position,

b. find, in terms of x, an expression for the tension in the rope at D. (3 marks)

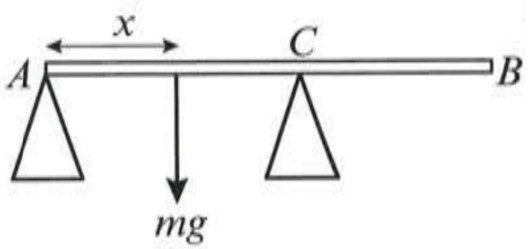
The rope at D will break if the tension exceeds $10g$ N. The rope at C cannot break.

c. Work out the range of possible values of x. (3 marks)

4.4 Centres of mass

Practice Book, printed page 38.

1. A non-uniform plank AB, of length 3 m, sits horizontally on supports at A and C where $AC = \frac{3}{5}AB$. The reactions at A and C are equal and the centre of mass of the plank lies a distance of x m from A. Work out the distance x.



Hint

The plank is non-uniform so the centre of mass is not at the midpoint of AB.

2. A non-uniform rod AB, of length 8 m and weight 2 N, rests on two supports positioned at A and B. The centre of mass of the rod lies 5.6 m from A.

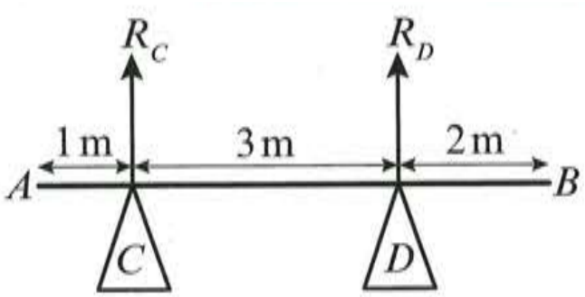
a. Draw a diagram to represent this situation. Include all the information you have been given.

b. Find the reactions acting on the two supports.

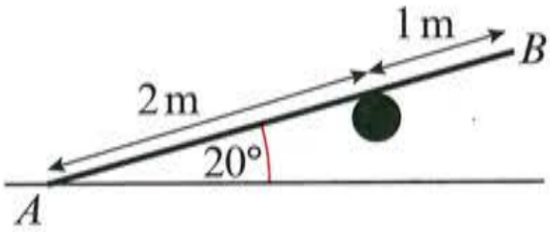
Hint

Take moments about A to find the reaction at B, and vice versa.

3. A non-uniform rod AB, of length 6 m, sits horizontally on two supports at C and D as shown in the diagram. The centre of mass of the rod lies 2 m from A. A mass, m, is loaded onto the rod at B, and the reactions at the two supports are now equal. Find, in terms of m, the mass of the rod.

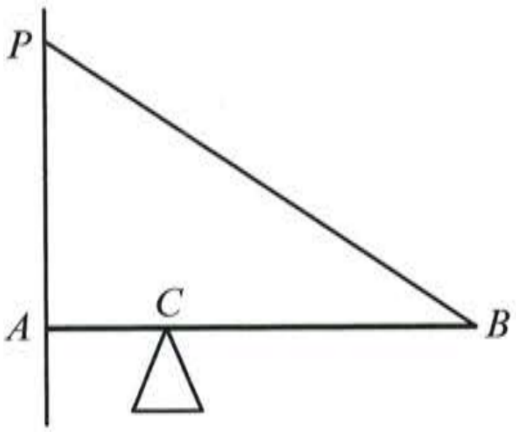


4. E/P A non-uniform rod AB, of length 3 m and weight 25 N, rests on a smooth peg that supports the rod 2 m from A. The end A rests on horizontal ground such that the rod is angled at 20° to the horizontal.
Given that the centre of mass of the rod lies 1.25 m from A, work out the magnitude of the reaction of the peg on the rod. (4 marks)

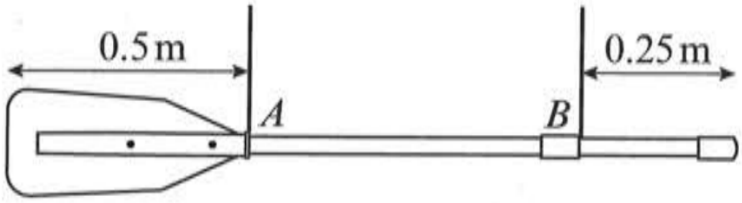


5. E/P A non-uniform platform AB is used to stand on to clean windows on a skyscraper. It is 4 m long and has a weight of 200 N. The platform is suspended at each end by vertical ropes so that it hangs in horizontal equilibrium. A window cleaner of weight 600 N stands on the platform 1.5 m from A. Given that the tension in the rope at A is 450 N, work out the distance of the centre of mass of the platform from B. (4 marks)

6. E/P A non-uniform rod AB, of length 3 m and weight 100 N, rests against a smooth vertical wall and on a support at the point C, where $AC = 1$ m. The rod is held in horizontal equilibrium by a rope connected to the rod at B, and to a fixed point P on the wall that lies 2 m vertically above A. Given that the tension in the rope is 100 N, work out the distance of the centre of mass of the rod from A. (4 marks)



7. E/P A rowing club has an oar hanging above its entrance door. The oar is suspended by two vertical ropes and held in the horizontal position as shown.



The oar is 1.5 m long and has a mass of 5 kg. It is modelled as a uniform rod and the ropes are modelled as light inextensible strings.

a. Using this model, calculate:

i. the tension in the rope at A

ii. the tension in the rope at B.

(4 marks)

b. Comment on the assumption that the oar can be modelled as a uniform rod. (1 mark)

The oar is now modelled as a non-uniform rod. The tension in the rope at A is four times the tension in the rope at B.

c. Work out the distance of the centre of mass of the oar from B. (3 marks)

4.5 Tilting

Practice Book, printed page 39.

1. A uniform beam AB, of mass 4 kg and length 5 m, rests horizontally on two supports positioned at C and D where AC is 1.25 m and DB is 2 m.



When a mass of m kg is placed at B, the beam is on the point of tipping about D.
Work out the value of m.

Hint

When the beam is on the point of tipping about D, the reaction at C is zero.

2. A uniform beam AB, of mass m kg and length 3 m, rests horizontally on two supports positioned at C and D, where AC is 1.25 m and DB is 0.75 m.

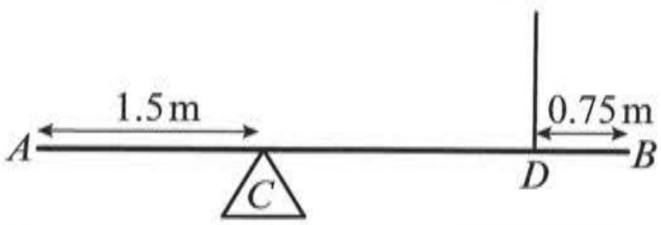


When a mass of 3 kg is placed 0.5 m from A, the beam is on the point of tipping about C.
Work out the value of m.

Hint

Take moments about C.

3. A uniform beam AB, of mass 20 kg and length 4 m, rests on a support positioned at C where AC is 1.5 m. It is held horizontal by a rope positioned vertically at D where DB is 0.75 m.



a. Find the tension in the rope.

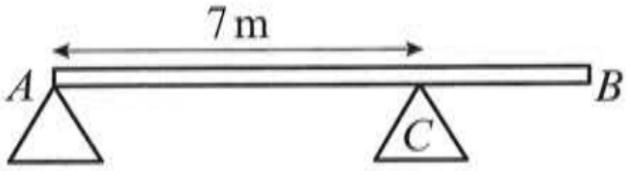
When a mass of 10 kg is placed x m from A, the beam is on the point of tipping about C.

b. Work out the value of x.

Hint

When the beam is on the point of tipping, the tension in the rope is zero.

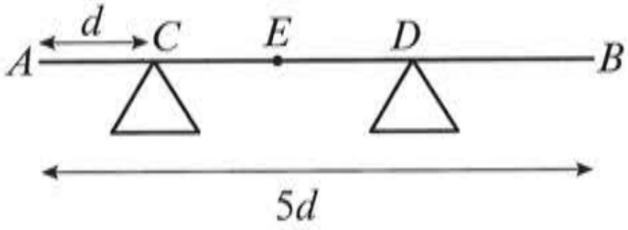
4. E/P A uniform plank AB, of mass 80 kg and length 10 m, rests on supports positioned at A and C where AC is 7 m.
A man walks along the plank to the point D where the plank starts to tip. Given that $AD = 9$ m, calculate the mass of the man. (4 marks)



5. E/P A non-uniform plank AB, of mass 120 kg and length 8 m, rests on supports positioned at A and C where AC is 5 m. The centre of mass of the plank lies $(3 + x)$ m from A. A mass of 150 kg is placed x m from C causing the plank to be on the point of tipping. Work out the value of x. (5 marks)

6. E/P A uniform plank AB, of length 6 m, hangs in equilibrium in a horizontal position from two vertical ropes attached at C and D where AC is 2 m and DB is 1 m. When a weight of 300 N is attached at A, the plank is on the point of tipping.
The weight at A is removed and a different weight, W N, is attached at B. Given that the plank is on the point of tipping, work out the value of W. (6 marks)

7. E/P A non-uniform rod AB, of mass $2m$ and length $5d$, rests horizontally in equilibrium on supports at C and D where $AC = d$ and $DB = 2d$. The centre of mass of the rod is at the point E. A particle of mass m is placed on the rod at B causing the rod to be on the point of tipping about D.

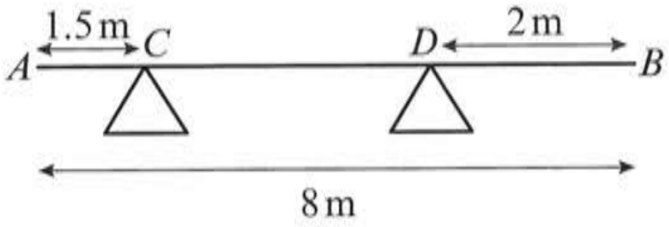


a. Show that $ED = d$ (2 marks)

The particle is moved from B to the midpoint of the rod and the rod remains in equilibrium.

b. Find, in terms of m, the magnitude of the normal reaction between the support at D and the rod. (2 marks)

8. E/P A uniform rod AB of length 8 m and mass m kg rests horizontally in equilibrium on supports at C and D where $AC = 1.5$ m and $DB = 2$ m. A mass of 1.5 kg is placed on the rod 0.5 m from A, causing the rod to be on the point of tipping about C.



a. Work out the mass of the rod, m. (2 marks)

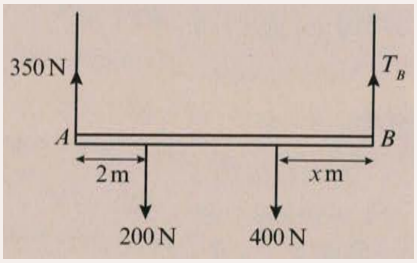
The 1.5 kg mass is now moved to a position between D and B, at a distance of x m from D. Given that the rod remains in equilibrium,

b. find the range of possible values of x. (2 marks)

Problem solving: Set A

Bronze

A non-uniform platform AB is 6 m long and has a weight of 400 N. Its centre of mass lies x m from B. It is suspended at each end by ropes so that it is horizontal. When a weight of 200 N is placed on the platform 2 m from A, the tension in the rope at A is 350 N.
Work out the value of x. (3 marks)



Silver

A non-uniform platform AB is 10 m long and has a weight of 900 N. It is suspended at each end by ropes so that it is horizontal. A weight of 600 N is placed on the platform 4 m from B. The tension in the rope at A is twice the tension in the rope at B.
Find:

a. the tension in each rope (2 marks)

b. the distance of the centre of mass of the platform from A. (2 marks)

Gold

A non-uniform platform AB is x m long and has a weight of W N. It is suspended at each end by ropes so that it is horizontal. A weight of $0.8W$ N is placed on the platform $0.25x$ m from B. The tension in the rope at A is half the tension in the rope at B.
Calculate the distance of the centre of mass of the platform from A. (4 marks)

Problem solving: Set B

Bronze

A uniform plank AB, of weight 400 N and length 20 m, rests on supports positioned at C and D where AC is 5 m and DB is 8 m. A 500 N weight is placed on the plank at a distance x m from A.
Given that the plank remains in horizontal equilibrium, find the range of possible values of x. (5 marks)

Silver

A uniform plank AB, of weight 400 N and length 15 m, rests on supports positioned at C and D where AC is 5 m and BD is 2.5 m. A weight, W N, is placed 1 m from A and the plank is on the point of tilting about C.
Work out the value of W. (4 marks)

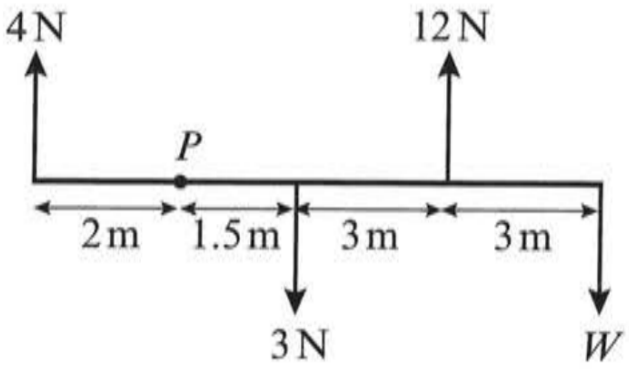
Gold

A uniform plank AB, of weight 400 N and length 6 m, rests on supports positioned at C and D where AC is x m and BD is $2x$ m. A weight, W N, is placed x m from B causing the plank to start to tilt.
Show that $x = \dfrac{1200}{W + 800}$ (4 marks)

Selected exam question bank questions

Question 21

The diagram shows the forces acting on a light rod. Given that the resultant moment acting about P is 12 Nm in the clockwise direction, work out the magnitude of force W. (2 marks)

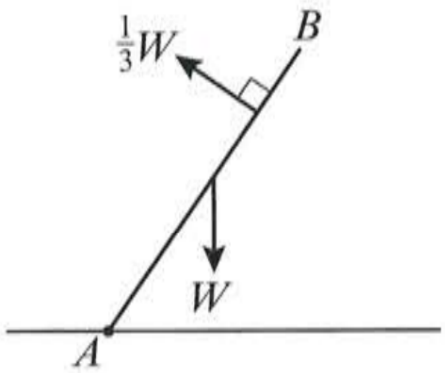


Question 24

A uniform rod AB, of length l m and weight W N, is hinged at A.

It is held in equilibrium by a force of $\frac{1}{3}W$ acting perpendicularly to the rod $\frac{1}{4}l$ m from B.

Work out the angle that the rod makes with the horizontal. (3 marks)

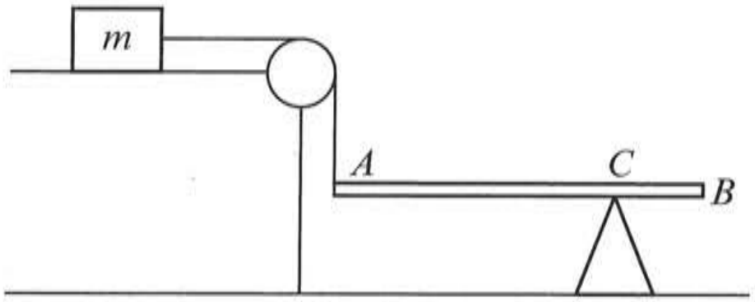


Question 28

A non-uniform rod AB, of length 16 m and weight 100 N, rests on two supports positioned at C and D. Support C lies 1 m from A and support D lies 6 m from B. A weight of 50 N is placed on the rod at B causing it to be on the point of tipping. Work out the distance of the centre of mass of the rod from A. (3 marks)

Question 31

A beam AB, of length l and mass M, is attached at A to one end of a rope that passes over a smooth pulley. The other end of the rope is attached to a box of mass m that lies on a rough horizontal table. The coefficient of friction between the box and the table is μ. The beam rests in a horizontal position on a support at C, where $CB = 0.25l$.



By modelling the box as a particle, the beam as a uniform rod and the rope as light and inextensible, show that $M \leq k\mu m$, where k is a constant to be determined. (4 marks)

Question 37

A uniform beam AB has mass 10 kg and length 6 m. The beam rests in equilibrium in a horizontal position, resting on two smooth supports. One support is at end A, and the other is at a point C on the beam, where $AC = 4$ m.

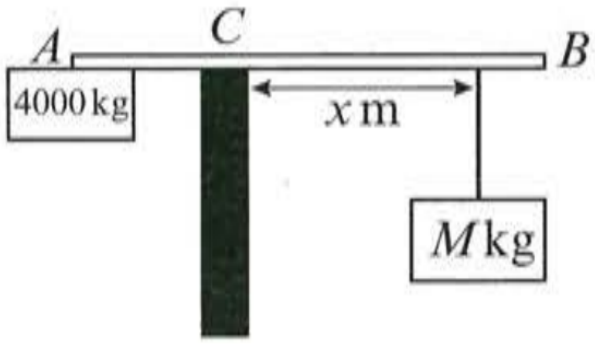
a. Find the reaction of the support on the beam at C. (3 marks)

A mass of 40 kg is placed on the beam at the point D. The beam remains in equilibrium and the reactions on the beam at A and C are now equal.

b. Find the distance AD. (7 marks)

Question 41

The beam of a crane is modelled as a uniform rod AB, of length 25 m and mass 5000 kg, resting in horizontal equilibrium. The beam is supported by a tower at C, where $AC = 10$ m. A counterbalance mass of 4000 kg is placed at A and a load of mass M kg is hung from a wire placed at a distance of x m from the supporting tower, where $x \geq 6$.

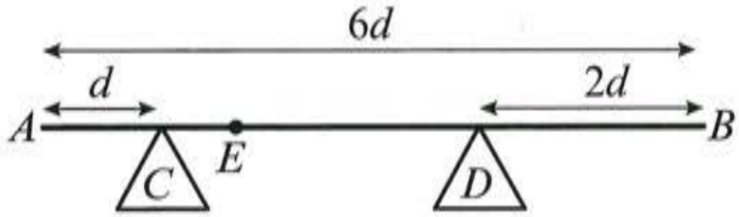


a. Determine the range of loads that can be lifted by the crane. (6 marks)

b. Criticise the model in relation to the beam of the crane. (1 mark)

Question 50

A non-uniform rod AB, of mass m and length $6d$, rests horizontally in equilibrium on two supports at C and D, where $AC = d$ and $DB = 2d$, as shown. The centre of mass of the rod is at the point E. A particle of mass m is placed on the rod at B and the rod is on the point of tipping about D.



a. Show that $AE = 2d$. (4 marks)

The particle is moved from B to the midpoint of the rod and the rod remains in equilibrium.

b. Find the magnitude of the normal reaction between the support at D and the rod. (5 marks)

Answers

4.1 Moments

1.
   a. 42 Nm in the anticlockwise direction
   b. 2.4 Nm in the anticlockwise direction
   c. 48 Nm in the clockwise direction
2.
   a. 7.05 Nm (3 s.f.) in the clockwise direction
   b. 77.3 Nm (3 s.f.) in the anticlockwise direction
   c. 4.2 Nm in the clockwise direction
3. 0.3 m
4.
   a. 80 N cm in the clockwise direction
   b. 80 N cm in the anticlockwise direction
5.
   a. 1 Nm in the anticlockwise direction
   b. 2 Nm in the anticlockwise direction
6. 30°
7. 0.4 m

4.2 Resultant moments

1. 5 Nm in the anticlockwise direction
2. 34 Nm in the anticlockwise direction
3. 0.161 Nm (3 s.f.) in the anticlockwise direction
4. 1 m
5. $\frac{19}{6}$ N
6. $\frac{23}{6}$ N
7.
   a. 312 kNm (3 s.f.) anticlockwise
   b. Assumptions include the barge is modelled as a rectangular lamina; direction in which the forces act are constant; forces are constant; resistance due to air/water is ignored.
8. Taking moments about P, the clockwise moment is

$$4F\cos 45^\circ - 5000\cos 45^\circ$$

   In order for the door to move, the clockwise moment must be greater than 0, so

$$4F\cos 45^\circ - 5000\cos 45^\circ > 0$$

   So $F > 1250$ N as required

4.3 Equilibrium

1.
   a. 50 N
   b. 100 N
2. $R_1 = \frac{50}{3}$ N, $R_2 = \frac{250}{3}$ N
3. 60°
4. $T_A = \frac{1}{3}mg$ N and $T_C = \frac{2}{3}mg$ N
5. 2.75 m

6. Taking moments about D:

$$\begin{aligned} m_A g \times 0.6l + Mg \times 0.1l &= m_R g \times 0.2l \\ 0.6m_A + 0.1M &= 0.2m_R \\ 6m_A + M &= 2m_R \\ M &= 2(m_R - 3m_A) \end{aligned}$$

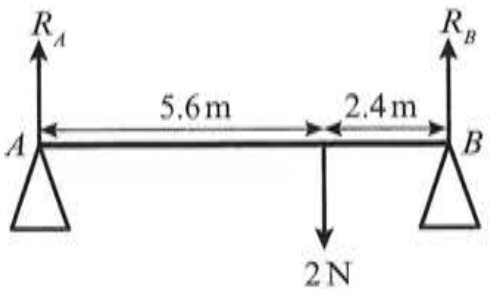
7. $10g\sqrt{2}$ N

8.
- **a.**
  - **i.** $\frac{10}{3}g$ N
  - **ii.** $\frac{20}{3}g$ N
- **b.** $\frac{5}{3}g(x+4)$ N
- **c.** $0 \leq x \leq 2$ m

4.4 Centres of mass

1. 0.9 m

2.
- **a.**



- **b.** $R_A = 0.6$ N, $R_B = 1.4$ N

3. $7m$

4. 14.7 N (3 s.f.)

5. 1.5 m

6. 2.11 m (3 s.f.)

7.
- **a.**
  - **i.** $\frac{10}{3}g$ N
  - **ii.** $\frac{5}{3}g$ N
- **b.** Mass of the oar is unlikely to pass through the midpoint of the oar.
- **c.** 0.6 m

4.5 Tilting

1. 1

2. 9

3.
- **a.** $\frac{40g}{7}$
- **b.** 0.5

4. 80 kg

5. $\frac{24}{27}$

6. 1200

7.

a. Take moments about D:

$$2mg \times ED = mg \times 2d$$

Simplify to $ED = d$.

b. $1.75mg$

8.

a. 0.6 kg

b. $0 < x \leq 0.8$

Problem solving: Set A

Bronze 3.25

Silver

a. $T_A = 1000$ N, $T_B = 500$ N

b. $\frac{14}{9}$ m

Gold $0.6x$ m

Problem solving: Set B

Bronze $1 \leq x \leq 13.6$

Silver 250

Gold Take moments about D:

$$\begin{aligned} 400(3 - 2x) &= Wx \\ 1200 - 800x &= Wx \\ (W + 800)x &= 1200 \\ x &= \frac{1200}{W + 800} \end{aligned}$$

Selected exam question bank answers

Question 21 7.13 N (3 s.f.)

Question 24 60°

Question 28 7 m

Question 31 Take moments about C to find $T = \frac{1}{3}Mg$.

At point of slipping, $T = \mu R = \mu mg$.

Equate values of T to find $M = 3\mu m$.

Question 37

a. $7.5g$ N

b. 1.75 m

Question 41

a. $1830 \leq M \leq 4580$ kg (3 s.f.)

b. The beam is unlikely to be uniform, or the beam does not rest freely on the tower.

Question 50

a. Let distance CE be x
Taking moments about D:

$$\begin{aligned}(3d - x)mg &= 2dmg \\ \Rightarrow 3d - x &= 2d \\ \Rightarrow x &= d \text{ and so } AE = 2d\end{aligned}$$

b. mg N